

IED SCOUT – VIRTUAL MACHINE

Objective

To configure a virtual machine so that **IED Scout** can communicate with a relay device over the network.

Prerequisites

- Virtual Machine software installed (e.g., VMware / VirtualBox)
 - IED Scout installed in the **Guest Machine**
 - Relay connected to the **Host Machine network**
 - Ethernet connection available
-

Basic Concept

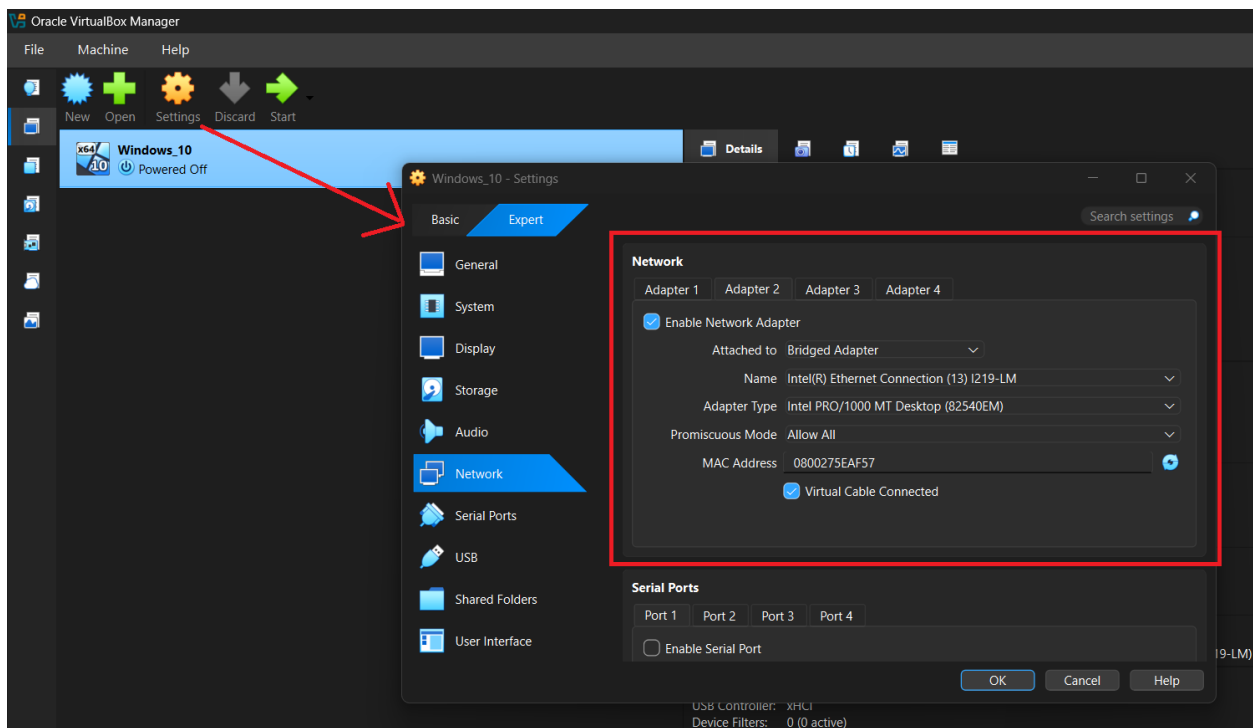
- The **Host System** is your main physical computer
 - The **Guest System** is the virtual machine running inside it
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- ➔ The goal is to make the **Guest Machine behave like a separate PC** on the same network as the relay.
- ➔ Follow the below steps to configure the settings.

Step 1: Configure VirtualBox Network Settings

1. Open **VirtualBox**.
2. Select the virtual machine and **discard the current machine state** (if applicable).
3. Navigate to:
Settings → **Expert Mode** → **Network** → **Adapter Settings**
4. Disable all network adapters and **enable only one adapter**.
5. Configure the enabled adapter as per the settings shown in the images below.

Note

Ensure that your VirtualBox network settings **exactly match** the configuration shown in the reference images.



Step 2: Configure IP Address Settings

1. Configure the IP address for the following devices:
 - **Host System**
 - **Guest System (Virtual Machine)**
 - **Relay**
2. Ensure that all IP addresses are in the **same network range (same subnet)**.
3. Make sure the IP address of the **Host System and Guest System are not the same** (each device must have a unique IP address).

Example (for clarity)

- Host System: 192.168.1.10
- Guest System: 192.168.1.11
- Relay: 192.168.1.12

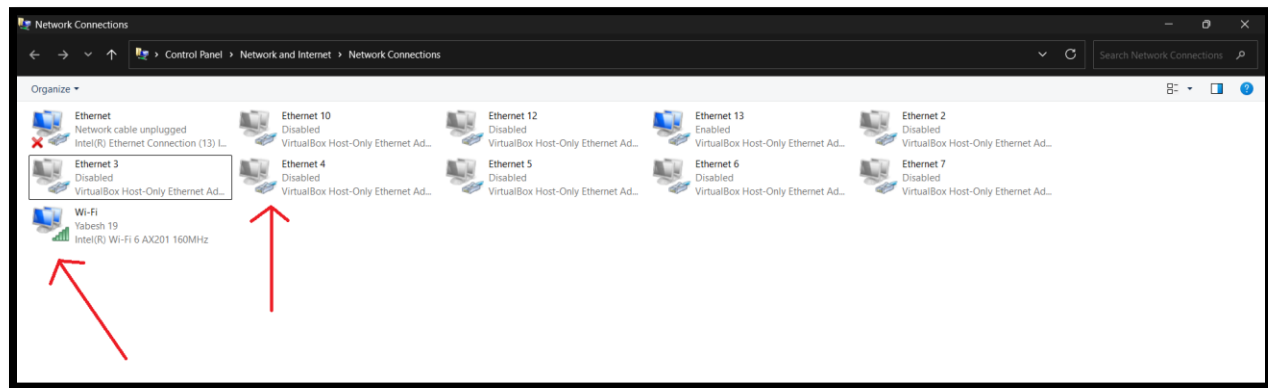
(All are in the same network: 192.168.1.x)

Step 3: Disable Unnecessary Network Adapters

1. Disable the following network adapters on the **Host System**:
 - **Host-Only Ethernet Adapter**
 - **Wi-Fi Adapter**
2. To disable:
 - Go to **Network Settings** on the Host System
 - Locate the adapters
 - Right-click and select **Disable**
3. Disable all **Host only adapter** of Virtual Machine on the Host system

Note

Only the required network adapter (used for relay communication) should remain enabled to avoid network conflicts.

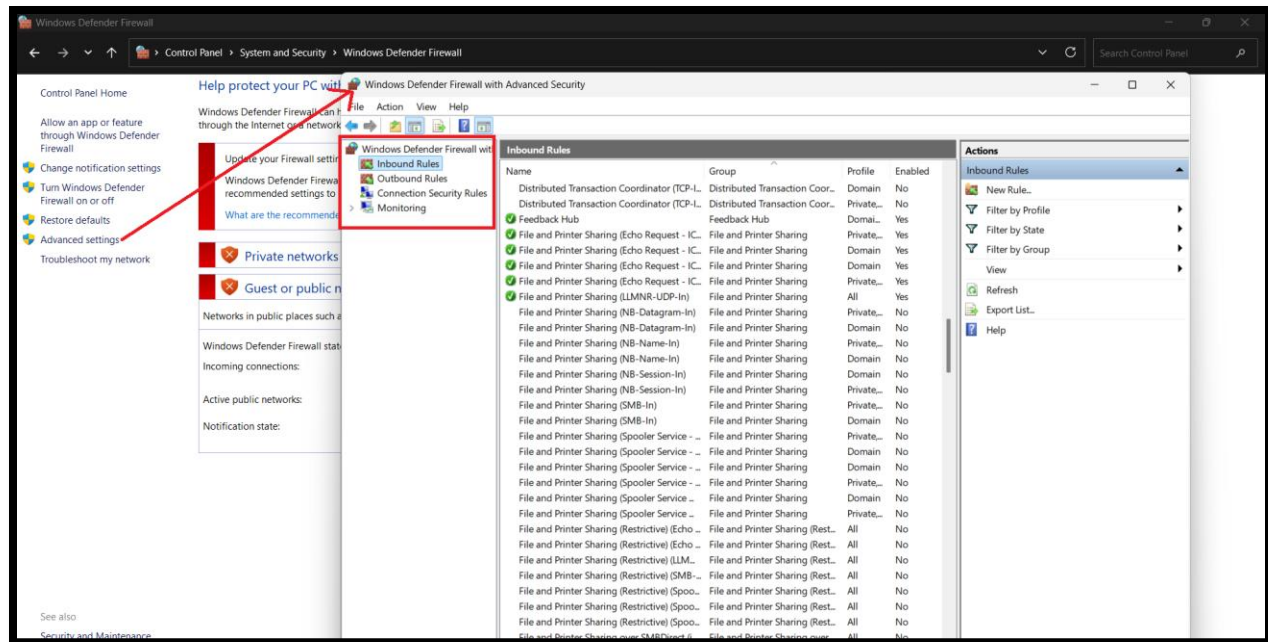


Step 4: Configure Firewall Settings

1. Perform the following settings on **both the Host System and Guest System**.
2. Turn **OFF the Windows Firewall** (temporarily, if required).
3. Navigate to the following path:
Control Panel → System and Security → Windows Defender Firewall → Advanced Settings
4. In the Advanced Settings window, locate:
 - **Inbound Rules**
 - **Outbound Rules**
5. Enable the required rules under **Private Profile**, such as:
 - **File and Printer Sharing (Echo Request – ICMPv4-In)**
 - Other necessary communication-related rules

Important Note

Ensure that firewall settings allow communication between the **Host, Guest, and Relay** systems



Step 5: Reset the Network Adapter

1. If the relay is **not reachable (ping not working)** from the Guest Machine, reset the network adapter on the **Host System**.
2. You can reset the adapter in two ways:

Method 1: Manual Reset

- Go to **Network Settings** on the Host System
- Locate the **Ethernet Adapter**
- Right-click → **Disable**
- Wait a few seconds → Right-click → **Enable**

Method 2: Using Command Prompt (Administrator)

Resetting the Network Adapter

1. Check available network interfaces:

```
>> netsh interface show interface
```

2. Disable the adapter:

```
>> netsh interface set interface "Ethernet" disable
```

3. Enable the adapter again:

```
>> netsh interface set interface "Ethernet" enable
```

Resetting the ARP Table

1. View the current ARP table:

```
>> arp -a
```

2. Clear the entire ARP table:

```
>> arp -d *
```

3. Delete a specific entry (example):

```
>> arp -d 192.168.1.10
```

Note

Resetting the adapter and clearing the ARP table helps resolve communication issues between the **Guest Machine and Relay**.

Step 6: Final Troubleshooting (Last Line of Defense)

1. If communication is still not established, **restart both the Host System and Guest System**.

Note

Restarting both systems can resolve temporary network or configuration issues that are not cleared by adapter reset.